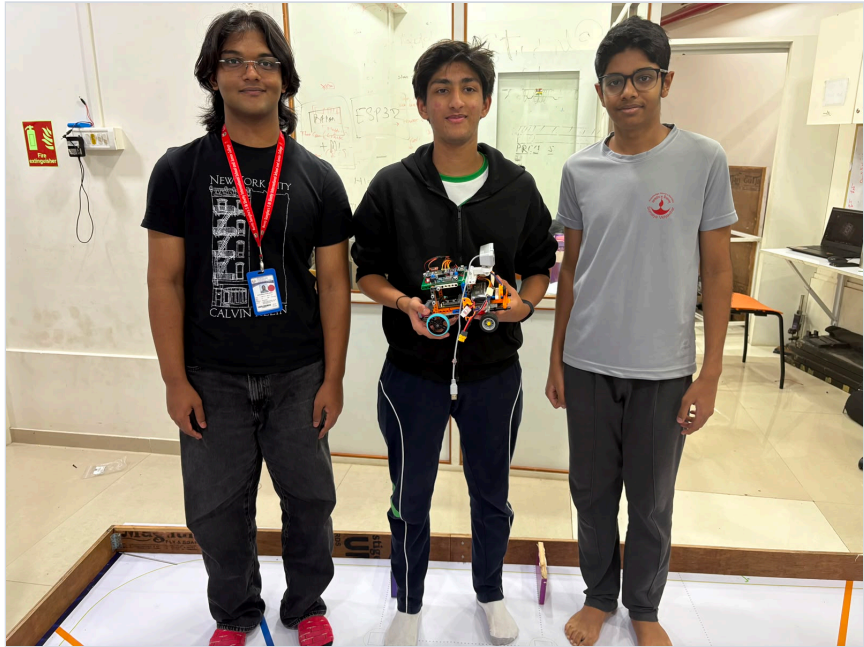


WRO FUTURE ENGINEERS 2026

FINAL ENGINEERING JOURNAL

TEAM TED DRIVE



Formal team photograph with the final competition vehicle

Team members	Ethan Fernandes — Software · Tejas Sirikonda — Mechanical · Diaan — Electronics
Coach	Amey Chavan (OMOTEC)
Competition	WRO Future Engineers 2026 · India Nationals, Hyderabad
Competition dates	26–28 August 2026

Repository: github.com/teddriveomo/wro-2026-future-engineers

Prepared 24 August 2026 (rev 3) against WRO 2026 General Rules, Appendix C (Engineering Journal evaluation).

01 Project Snapshot

A concise map of the final system, evidence and current validation state

Our vehicle is a fully autonomous four-wheel car built around a rigid LEGO Technic hybrid frame, single-servo Ackermann steering and a rear-drive differential. An ESP32 performs the time-critical sensing and steering loop, while a Raspberry Pi 5 handles pillar perception. The architecture was intentionally split so that a slow or failed vision process cannot stall steering.

Item	Final / current state
Final robot mass	785 g, including competition electronics and battery
Open Challenge	Implemented and tested; latest reported three-lap result: 24 s
Obstacle Challenge	Competition stack committed at <code>src/Round 2/competition/</code> : ESP32 firmware (<code>test.cpp</code> — PID heading hold, 14-state machine, reverse-arc corners, PAUSE→PASS→SIDE→YAW-BACK pass sequence) + Pi detector (<code>detector.py</code> — 2-class green/red ONNX, 5-of-7 vote, 7-column offset LUT). Earlier stacks (3-class detector line, CIELab picker) retained in-repo as evidence. Final timed track test pending.
Parking	Parking-lot start / exit implemented in the competition firmware (boot S-curve: reverse → 45° swing → straight leg → align, heading-verified). End-of-run parallel park-in: shimmy controller written but not yet run (branch <code>feat/round2-park</code>); magenta bay-marker detection exists in the 3-class detector line.
Main power switch	Installed on final vehicle (with separate start button — WRO two-button start sequence)
Steering	MG90 metal-gear servo; Ackermann; 64–136° travel about a 106° centre
Range / heading	3× TF-Luna through PCA9548A + BNO055 absolute yaw
Compute	ESP32 for motion; Raspberry Pi 5 for vision
Drive	N20 DC motor → TB6612FNG → 20T:28T → LEGO differential

WRO documentation target. Appendix C awards up to 6 points each (scored 0/2/4/6) for Mobility, Power/Sensors, Software/Obstacle Strategy, Systems Thinking, and Reproducibility/GitHub Quality. This journal is organised in exactly those five areas and emphasises decisions, tests, iterations and rebuildability rather than appearance alone.

Section	What the judges can find
1. Mobility & Mechanical Design	Chassis, drivetrain, steering geometry, torque/traction/speed derivation, CAD of printed parts, mechanical iteration, final mass
2. Power & Sensor Architecture	Power tree, sensor selection, placement, I2C architecture, calibration and failure handling
3. Software & Obstacle Strategy	Control-design rationale (P vs PID), Open control flow, firmware logic excerpt, Round 2 state machine, perception pipeline with code, tuning evidence
4. Systems Thinking & Engineering Decisions	Constraints, trade-offs, rejected alternatives, iteration cycles and risk mitigation
5. Reproducibility & GitHub Quality	Repository structure, code/CAD/wiring/video evidence, environment and rebuild workflow
Testing & Final Status	Latest Round 1 result, pending Round 2/parking, pre-competition verification items

02 Engineering Process

The journal records how the design changed, not only what the robot looks like now



Informal engineering-session photograph: reviewing the field, pillar placement and robot behaviour during development

The project progressed through repeated plan–build–test–improve cycles. Several ideas that looked correct on paper were changed after measured failures. We deliberately kept those reversals in the repository because they explain why the final system exists in its present form.

Engineering iteration timeline

Date	Milestone
26 Jun	Initial motion firmware — heading hold + corner detection
19 Jul	HSV pipeline measured; failure modes quantified
26 Jul	Clean dataset split rebuilt after leakage audit; NanoDet line adopted
5 Aug	Field stack reworked; repo restructured to the five-criterion layout
9–10 Aug	Chassis stiffness redesign validated; steering retune + correction-sign fix
11–12 Aug	CAD collected, videos recorded/uploaded, magenta class added, documentation snapshot
19 Aug	3-class YOLO26n v2 retrained on the leakage-fixed split (Colab, 80 epochs)
21 Aug	Round-1-lineage obstacle firmware + Pi vision sender committed and merged to main
24 Aug	Competition Round-2 stack committed (src/Round 2/competition/); capture tooling merged; final journal rev 3

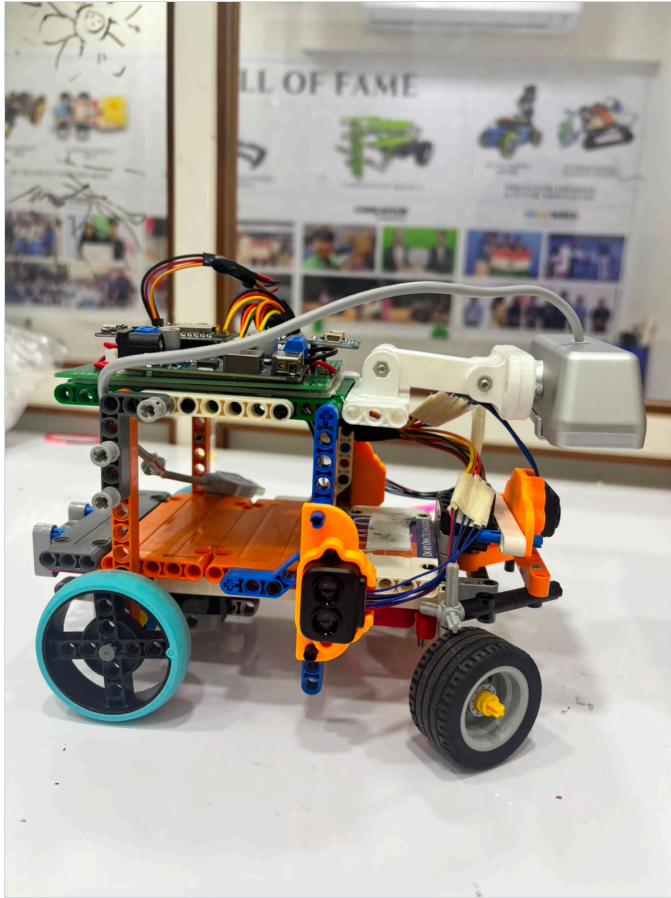
The strongest iteration examples

- Corner detection changed from a level test to a rising-edge event because a level condition repeatedly stepped the target heading and could make the robot spin.
- The first LEGO frame flexed under drivetrain load and shed wheels. Instead of immediately replacing it with a printed chassis, we redesigned the LEGO space-frame to be much stiffer; later sustained runs no longer lost wheels.

- The first neural pillar detector performed well on the clean validation split but transferred poorly to real field footage. We fell back to a per-venue calibrated CIELab colour picker, retrained a 3-class YOLO26n on a leakage-fixed split, and finally converged on the 2-class grid/pause-pass competition stack — **every stage is kept in the repository as evidence** (§08–§09).
- The steering correction **sign was inverted** until the 10 Aug retune — the servo initially fought the heading error instead of nulling it. Caught on the mat, fixed in one commit, and documented rather than hidden (§07).
- Sensor power routing was reworked so the MG90 servo receives 5 V from the supply rail while the ESP32 provides only the control signal, reducing a previously identified failure path.
- A dedicated main power switch has now been installed on the final vehicle, completing the WRO start sequence alongside the separate start button.

03 Criterion 1 — Mobility & Mechanical Design

How the chassis, drive and steering were chosen, measured and refined



Side view: electronics stack, rear drive wheel and front steering



Opposite side: camera mount, LiDAR and drivetrain packaging

Final mechanical architecture

Subsystem	Final design	Engineering reason
Chassis	Rigid LEGO Technic hybrid space-frame with printed mounts	Rebuildable, light and sufficiently rigid after the flex-induced wheel-loss failure was diagnosed.
Steering	One front steered axle, single MG90 metal-gear servo, Ackermann linkage	Complies with the non-skid steering requirement and gives a predictable geometric steering response.
Drive	Rear axle driven by one N20 motor through TB6612FNG and LEGO differential	Both driven wheels remain mechanically connected; the differential allows wheel-speed difference during cornering.
External reduction	20-tooth pinion → 28-tooth crown = 5:7 (1.4:1), counted on the vehicle 11 Aug	Trades some speed for torque/control without needing a drivetrain rebuild.
Wheels	Rear 55.6×14 mm; front 41×21 mm	Larger driven rear wheels improve distance per revolution; smaller front wheels suit the steering geometry.
Mass	785 g final	Only 52.3% of the 1.5 kg limit, leaving 715 g of margin.

Printed parts — CAD (designed in Fusion 360, exported STL, in models/)



N20 motor clamp — LEGO-pitch holes Servo horn beam — Ackermann link TF-Luna enclosure — LEGO-pitch

Each part solves a LEGO↔commodity-hardware interface: the clamp holds the round N20 body at LEGO pitch, the horn beam converts the servo horn into a Technic-pin link, and the enclosure fixes each TF-Luna's aim height and yaw. The LEGO chassis itself is captured as chassis_lego_current.lxfml plus step-by-step build instructions (PDF) in models/chassis/.

Torque, traction and acceleration (derived; datasheet-class motor figures, labelled accordingly)

Speed answers "how fast can it go" — this answers "can it get there, and what stops it first." The motor is a GA12-N20-600 class unit: 12 V rated, 600 rpm no-load, 1:50 gearbox, 0.18 kg·cm rated / 0.65 kg·cm stall torque. The external stage multiplies motor torque by $R_{ext} = 1.4$ (20T→28T, counted on the vehicle):

	Motor (datasheet-class)	× 1.4 → wheel	Tractive force at r = 27.8 mm
Rated	0.18 kg·cm = 0.0177 N·m	0.0247 N·m	0.89 N
Stall	0.65 kg·cm = 0.0637 N·m	0.0892 N·m	3.21 N

What stops the vehicle first — and it is genuinely close. The traction limit is $\mu \cdot N_{rear}$. At the measured 785 g with 50–60% of weight on the driven axle and rubber-on-painted-mat $\mu \approx 0.5\text{--}0.9$, the traction ceiling spans roughly **1.9–4.2 N**. Against a stall tractive force of **3.21 N** the design sits *on the boundary*: at the slick end the wheels break traction before the motor stalls; at the grippy end the motor stalls first. The distinction is not academic — it decides whether a wall strike stalls the motor (current spike, TB6612 thermal event) or spins the wheels (no spike, but no recovery either). A front/rear axle-load measurement settles the regime and is listed as an open verification (§13).

Acceleration. Motor torque falls roughly linearly from stall toward rated as speed rises; a mid-ramp average of ~2 N on 0.785 kg gives $\approx 2.5\text{ m/s}^2$, so the vehicle reaches its derived 1.15 m/s ceiling in well under a second and under half a corridor width — it is at speed before the first corner, which is why the final run-on behaves as a distance rather than a ramp.

Does the ratio give enough speed? The driving corridor is ~1000 mm inside a 3200 mm mat, so a centre-line lap is roughly $4 \times 2.2\text{ m} \approx 8.8\text{ m}$ and three laps $\approx 26\text{ m}$ — about 23 s of pure driving at the 1.15 m/s ceiling before corner slowdowns. The 5:7 stage is therefore not the binding constraint on round time, which is the justification for leaving it rather than re-cutting for speed.

04 Steering Control Design & Mechanical Testing

Why the controller is proportional-only, and the test evidence that changed the physical design

Control design — Round 1: PID evaluated, P-only selected

The steering loop converts wrapped heading error ($\pm 180^\circ$) into a servo correction. We evaluated the full PID family against the specific structure of this task before choosing the final law:

Term	What it would do here	Decision
P	Correction proportional to heading error — restoring torque toward the target heading.	Kept. Gain = 1.0 servo-degree per degree of error; a 90° corner command drives the servo straight to full lock, so corners need no separate manoeuvre.
I	Accumulates error over time to remove steady-state offset.	Rejected. Our target heading is <i>deliberately stepped</i> by 90° at each corner — a commanded discontinuity. An integrator winds up across the entire transit of every corner and then discharges as overshoot on the next straight. The plant also has no steady-state error source that P + geometry does not already handle.
D	Damps by reacting to error rate.	Rejected. The BNO055 yaw is quantised; differentiating it amplifies quantisation noise into servo chatter, wearing the MG90 and adding jitter on straights.

The final law is **P = 1.0 deg/deg with a 2.0° deadband**, output clamped to the measured mechanical window $64\text{--}136^\circ$ about the 106° centre (asymmetric $-42^\circ/+30^\circ$). The deadband suppresses hunting inside the sensor's noise floor; the clamp guarantees no software state can command the linkage past a hard stop. Straight-line gain can then be tuned for stability rather than corner aggressiveness, because corners are executed by the target step saturating the P term — not by a separate turn routine.

Round 2 runs the full PID — and the difference is structural, not taste. The obstacle firmware (src/Round 2/competition/test.cpp) holds heading with **Kp = 1.5, Ki = 0.02, Kd = 0.15** on the wrapped $\pm 180^\circ$ error: P and I are gated by the heading deadband, the integral is hard-clamped at ± 50 (anti-windup), and the total correction is clamped to ± 20 servo-degrees. Round 2's disturbance profile is different from Round 1's — the pass sequence holds sustained steering offsets and re-join arcs rather than stepping the target, so a derivative term is justified damping the return to the lane, and a small, clamped integral trims residual bias during long offset holds. Both clamps bound each term's failure mode, which is exactly what makes I and D admissible here after being rejected for Round 1. Each round runs the controller its disturbance profile justifies.

Iteration evidence — the sign bug. Until the 10 Aug retune the correction *sign was inverted*: the servo steered into the error instead of nulling it, and the "centre" and limits were also stale ($45\text{--}135^\circ$ about 90° from an earlier linkage). The on-mat symptom — divergence instead of convergence — was traced in one session; the fix (sign flip, centre 106° , limits $64\text{--}136^\circ$, gain 1.0, deadband 2.0°) is a single documented commit. We kept the wrong version in history because it is the reason the current numbers exist.

Failure → diagnosis → redesign (chassis)

Stage	Observed evidence	Decision / result
Initial sustained drive	Wheels worked loose from the axles within seconds.	Treat wheel retention as a frame/stiffness problem, not only a wheel problem.

First proposed fix	A fully 3D-printed chassis was considered to add rigid hubs and retention.	Kept as a fallback, but not committed as the final answer.
Final fix	Ground-up redesign of the LEGO space-frame; tighter packaging and a lower, more rigid load path.	Two later runs of 28 s and 29 s continuous driving recorded no wheel loss; the printed chassis rebuild was cancelled.
Final configuration	Same stiff frame used for all later work; full electronics and camera mounted.	Final competition mass measured at 785 g.

Latest Open Challenge validation

Test	Result	Interpretation
Three-lap Open Challenge	24 s total	8 s per lap average; confirms complete three-lap operation on the latest reported test.
Automatic stop logic	12 corners observed, then final run-on and braking	Software counts 90° heading steps and stops after the third lap rather than relying on elapsed time alone.
Mass check	785 g	Large legal margin below the 1.5 kg limit and less load on steering / drivetrain.

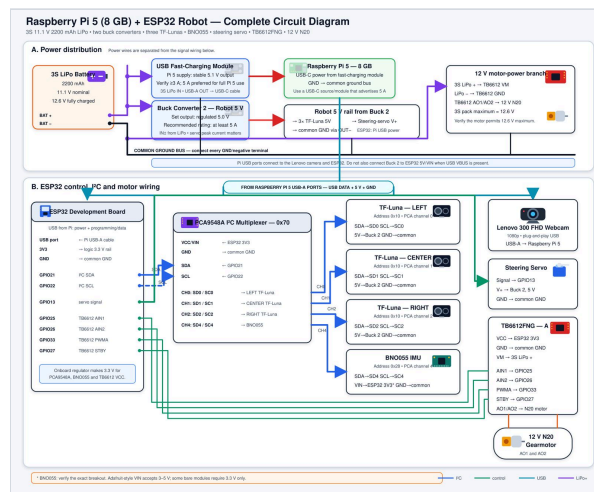
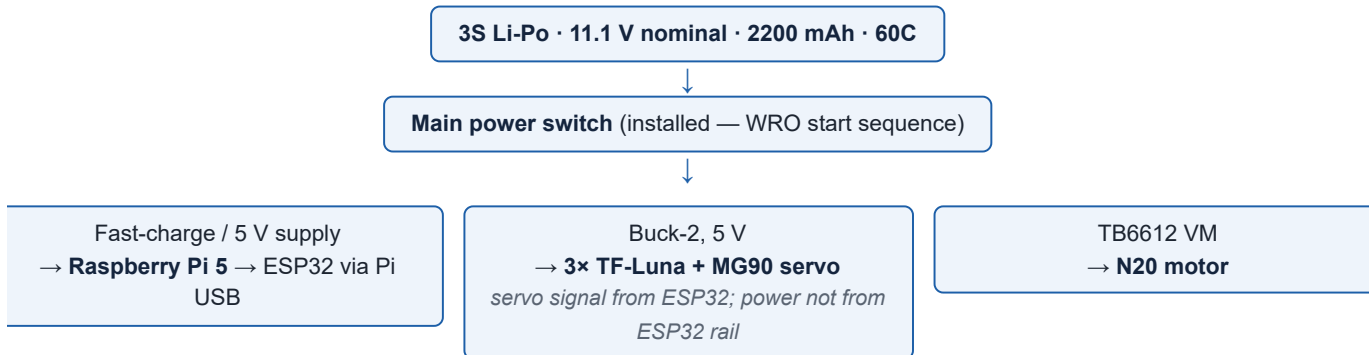
Cross-checking the 24 s result against the drivetrain model. 24 s over three laps means the average speed equals the driven lap length $\div$ 8 s. On the centre-line estimate (~ 8.8 m/lap) that would imply ~ 1.1 m/s — implausibly close to the 1.15 m/s *unloaded* ceiling — so the result is consistent only with the shorter inner line the controller actually drives after each corner (~ 6.5 – 7.5 m/lap), giving a loaded average of **0.8–0.95 m/s $\approx$ 70–83% of the no-load ceiling**. That is aggressive but physically coherent; a tape measurement of the driven line is listed as an outstanding verification (§13), and the loaded straight-line speed itself is re-measured by tape + timer before the rounds. This journal does not carry a straight-line speed number until that measurement exists.

The Open Challenge test is the strongest complete end-to-end validation currently available. The Obstacle Challenge and parallel-parking validation are kept separate in later sections because they are not yet complete; this journal does not convert unfinished work into claimed results.

05 Criterion 2 — Power & Sensor Architecture

How sensing and power are organised so that the system remains reliable and reproducible

As-built power architecture



Complete as-built circuit, photographed 11 Aug; drawn schematics (signal + power, v0.2) are versioned in schemes/

Power distribution decisions

- 3S Li-Po pack: 11.1 V nominal, 2200 mAh, 60C.
- Main power switch is installed in the final vehicle so the robot can be placed on the field switched OFF, then powered ON before the separate start command.
- The Raspberry Pi 5 is powered through the vehicle's regulated 5 V supply; the ESP32 is supplied from the Pi USB path.
- A separate 5 V buck rail powers the three TF-Lunas and MG90 servo. The servo control signal comes from the ESP32, but the servo is not powered from the ESP32 rail.
- The motor rail feeds TB6612FNG VM and the N20 separately from the 5 V electronics branch, reducing interaction between motor load and sensing/control.
- **Predicted power budget** (derived, versioned in docs/2_power_predicted_budget.md): cruise draw $\approx$ 0.95–1.5 A at the pack → 23–37 three-minute runs per charge on 1760 mAh usable. Perception (Pi + camera) costs $\approx$ 3× propulsion; the binding constraint is Pi 5 thermal throttling (82 °C onset), not pack capacity.

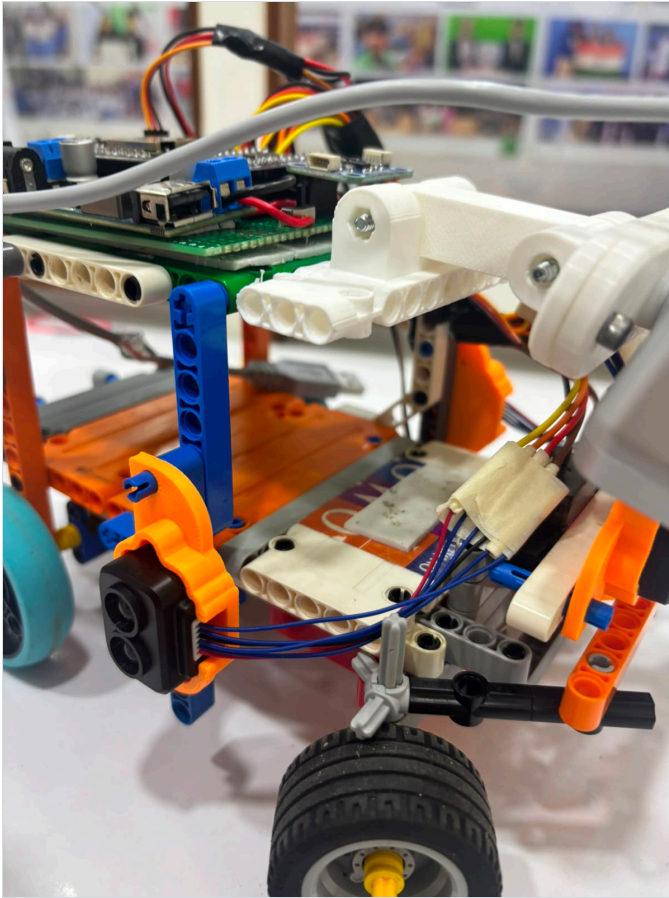
Sensor map and why each device exists

Device	Qty / bus	Role	Reason for selection
BNO055 IMU	1 · I ² C 0x28 · mux ch4	Absolute yaw / heading	Avoids three-lap drift associated with integrating gyro rate; heading targets are stepped by exact 90° increments.

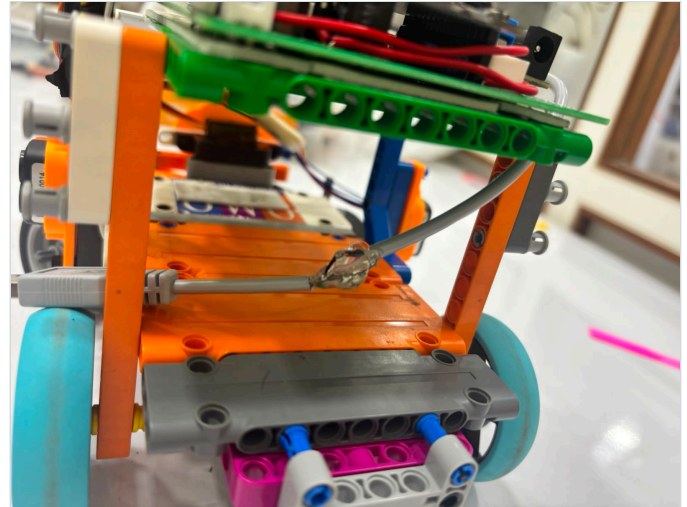
TF-Luna LiDAR	3 · I ² C 0x10 · mux cho/1/2	Left, centre, right distance	Narrow ~2° beam gives a sharper wall- opening edge than the much wider ultrasonic cone.
PCA9548A	1 · I ² C 0x70	I ² C channel multiplexer	All TF-Lunas share address 0x10, so each must be isolated on its own channel.
USB camera	1 · Raspberry Pi 5	Red / green / magenta visual detection	Colour + bearing for obstacles and parking markers.

06 Sensor Placement, Calibration & Failure Handling

The sensing layout was designed around field geometry and known failure modes



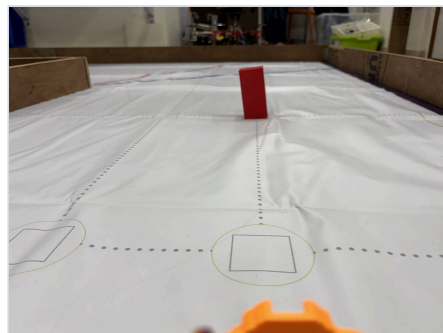
Front sensor + camera mount as built — TF-Luna aim height and camera pitch fixed by the printed mounts (\$03 CAD)



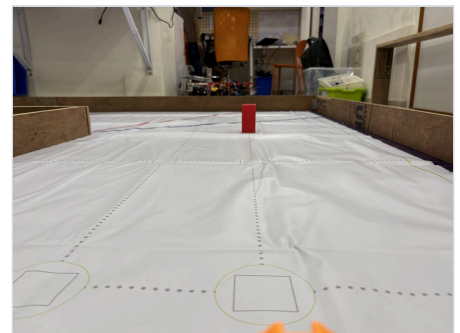
Chassis midsection: mux, wiring loom and load path



Camera POV — near pillar



Camera POV — mid range



Camera POV — far; sets the minimum detectable box height

Geometry and bus architecture

The side TF-Lunas watch for the track wall to disappear. A corner is not treated as "distance is large"; it is treated as the transition from wall-present to an opening over 150 cm. That distinction makes sensor placement and beam width more important than absolute range accuracy. The centre TF-Luna provides forward clearance; in Round 2 it is the corner trigger (reverse-arc at 20 cm).

All three TF-Lunas respond at I²C address 0x10, so direct parallel connection would cause an address collision. The PCA9548A solves this by selecting one channel at a time: left on channel 0, centre on 1, right on 2, and the BNO055 on channel 4. The ESP32 I²C bus uses GPIO21 (SDA) and GPIO22 (SCL).

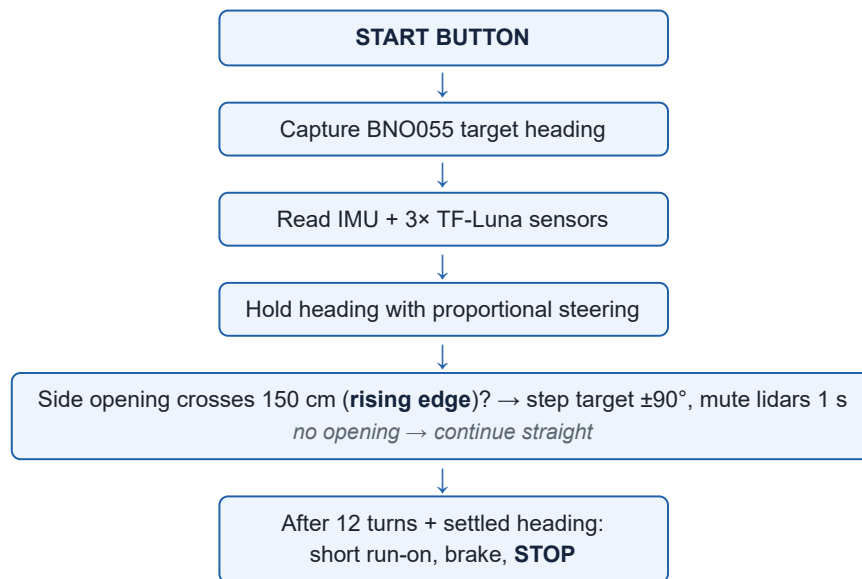
Calibration and noise controls

- BNO055 heading is captured at the start button, after the robot has been positioned on the field, so the reference frame follows the actual starting orientation.
- Round 1 accepts a corner only on a rising edge with a 1 s lidar mute; Round 2's corner trigger is the front LiDAR at 20 cm with a 1 s cooldown — in both designs a pillar briefly occluding a side sensor cannot fire a turn.
- The competition detector's class map is **locked in code** ({0: green, 1: red}); the 3-class line's map ({0: green, 1: red, 2: magenta}) is likewise locked, because the exported ONNX carries no name metadata — the constant is always the map in use.
- Temporal voting — 5-of-7 agreeing frames with a 70 px spatial lock before a pillar is confirmed, and a 3-frame CLEAR debounce — prevents a single bad frame from commanding or cancelling a pass.
- In the 3-class detector line, the magenta parking-marker path includes a dedicated chroma floor and morphological opening to separate marker colour from reflection bleed on the mat.

Failure-aware design. The repository explicitly treats lighting changes, sensor occlusion, I²C address collisions, motor magnetic interference and perception stalls as engineering risks. The two-processor split ensures a perception stall does not block the steering loop.

07 Criterion 3 — Software Architecture & Open Challenge

The motion controller is event-driven, heading-based and designed for repeatability



Open Challenge algorithm

- When the start button is pressed, capture the current BNO055 yaw as the target heading.
- Read the IMU and TF-Luna sensors continuously.
- Use proportional steering (\$04 control design) to reduce wrapped heading error while driving forward.
- Detect a corner only when a side distance crosses from wall-present to an opening greater than 150 cm.
- Step the target heading by +90° or −90° toward the opening; mute the lidars for 1 s so the opening cannot re-trigger while it remains visible.
- Count the completed 90° target-heading steps. After 12 turns, wait until heading is settled, perform the final short run-on, brake and stop.

Firmware logic — verbatim excerpt from `src/Round 1/round 1/round 1.ino` (tunables + rising-edge corner detector)

```

#define SERVO_MAX_ANGLE 136 // servo upper limit (deg)
#define SERVO_CENTER 106 // wheels-straight angle
#define STEER_GAIN 1 // servo deg per deg of heading error
#define OPENING_CM 150 // side reading above this = corner/opening
#define TURN_STEP_DEG 90 // heading change per detected opening
#define STEER_DEADBAND 2.0
#define LOOP_DELAY_MS 20
// ... inside loop(): heading always read, steering always live ...
bool lOpen = (l > OPENING_CM);
bool rOpen = (r > OPENING_CM);

bool turned = false;
if (lOpen && !leftWasOpen) { targetHeading -= TURN_STEP_DEG; turned = true; }
if (rOpen && !rightWasOpen) { targetHeading += TURN_STEP_DEG; turned = true; }

leftWasOpen = lOpen;
rightWasOpen = rOpen;

if (turned) lidarResumeAt = millis() + 1000; // corner taken -> mute lunas 1 s
// during the pause: no reads, no corner detection – but steering stays live
int servoAngle = steerToHeading(h, targetHeading);
  
```


Why the rising-edge corner detector matters

A threshold condition such as "if distance >150 cm, turn" remains true for many control cycles while the sensor is looking into the open corner. Without memory of the previous state, the target heading could be stepped repeatedly by 90°, producing a spin. The rising-edge design fires once at the instant the wall disappears (`!Open && !leftWasOpen`), then the 1 s lidar mute plus the stored wall state require a genuine new edge before another corner can be detected. Steering remains live throughout the mute — the P-controller saturating against the stepped target is what physically executes the corner.

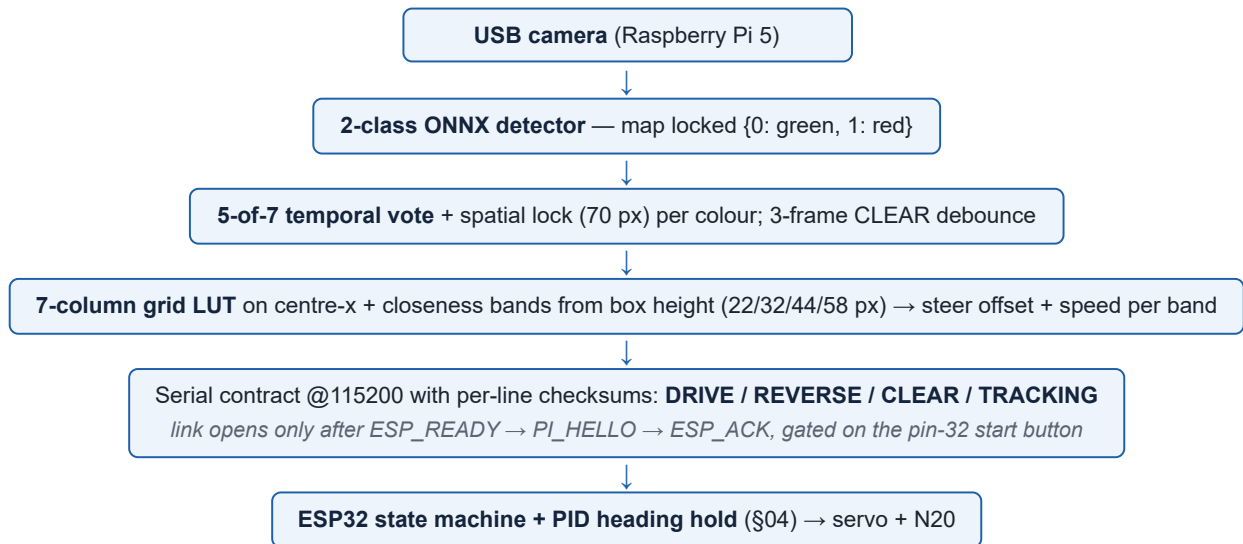
Latest measured outcome

The latest team-reported Open Challenge test completed all three laps in 24 seconds (analysed against the drivetrain model in §04). It demonstrates the interaction of chassis rigidity, drivetrain, heading control, corner detection, lap counting and autonomous stop logic rather than testing any one module in isolation.

o8 Obstacle Challenge Strategy

A staged pass sequence, a PID heading hold and a checksummed serial contract between the two processors

Perception → decision → actuation pipeline (competition stack, src/Round 2/competition/)



Implemented state logic (ESP32, 14 states)

State(s)	Purpose	Key behaviour / exit
PARK_WAIT → PARK_REVERSE → PARK_PHASE1 → PARK_STRAIGHT_LEG → PARK_PHASE2	Start inside the parking lot; S-curve exit at boot	Reverse out of the slot, 45° swing away, forward straight burst, counter-swing to align — heading-verified against the start reference, with an 8 s safety cap
DRIVING_STRAIGHT	PID heading hold and obstacle monitoring	Confirmed pillar → pass sequence; front TF-Luna under 20 cm → corner
PRE_TURN_REVERSE / TURNING	Reverse-arc 90° corner	If a far cube is still in view, IMU reverse on the cardinal first (650 ms, to 28 cm clearance), then the reverse-arc turn; target snapped to the cardinal grid; 1 s cooldown
OBSTACLE_AVOIDING (PAUSE → PASS)	Execute the pass on the commanded side	400 ms full stop at 45 px box height with pass steer held frozen, then wide pass (full 35° swerve floor) until the cube leaves the camera edge
PI_SIDE → PI_YAW_BACK	Re-join the lane after the cube is off-camera	Drive straight until the matching side LiDAR (red = left, green = right) sees the cube at ≤20 cm — or a 1.6 s cap — then a fixed 28° yaw back toward the lane
PI_REVERSE	Recover when pillar geometry becomes too close	Short 550 ms reverse bursts (max 3), then retry the pass — never latch
ROBOT_STOPPED	End after 12 counted turns (3 laps)	Terminal state

Obstacle perception and pass-side decision

The Pi runs a 2-class green/red ONNX detector. Each colour keeps a 7-frame history: a pillar is *confirmed* only on 5-of-7 agreeing frames with a 70 px spatial lock, and a CLEAR must persist 3 frames — a single bad frame can neither command a pass nor cancel one. The nearest confirmed box is placed on a 7-column grid over centre-x with closeness bands from box height; the LUT emits a steering offset (up to the full 35° swerve for both colours — an earlier on-robot LUT that peaked below full lock let the body clip the cube, which is why the floor is now 35°) and a band speed. Red passes right, green passes left. At 45 px box height the sequence inserts a deliberate 400 ms full stop with the pass steer frozen, then commits to the wide pass; after the cube leaves the frame the firmware goes straight until the *matching side LiDAR* confirms the cube alongside, then yaws back a fixed 28°. A segment allows at most 2 passes before the next corner.

Pi-side detection & pass-side decision — verbatim excerpts from `src/Round 2/competition/detector.py`

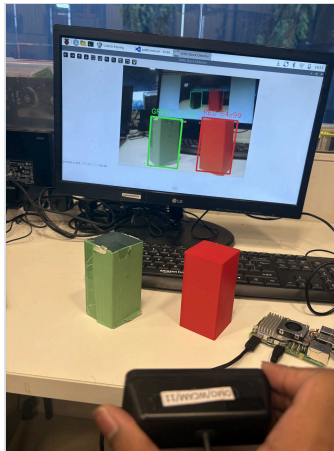
```
ONNX_MODEL_PATH = "best_ncnn.onnx"
CLASS_NAMES = {0: "green", 1: "red"}
VOTE_HISTORY = 7
MIN_VOTES = 5
CLEAR_HISTORY = 3
LOCK_MATCH_PX = 70
SWERVE_HEIGHT_PX = 45          # start the pass only when the cube is this tall
ALIGN_PAUSE_S = 0.40          # stop 400 ms at 45px, then swerve
GREEN_MIN_OFFSET = 35         # green pass-left needs full DIFF or the body clips
RED_MIN_OFFSET = 35           # red pass-right, same full swerve
# ... main loop: confirmed 5-of-7 per colour, nearest actionable box wins ...
cmd = (
    f"DRIVE,{primary_color},{cmd_info['speed']},{cmd_info['offset']},"
    f"{cmd_info['phase']},{active_box['height']}"
)
if cmd != last_sent or due:
    serial_write(cmd)
elif decision == "CLEAR" and clear_counter >= CLEAR_HISTORY:
    serial_write("CLEAR")
```

Model file: the weights sit next to the script under the name in `ONNX_MODEL_PATH`; the 3-class v2 weights versioned in `../detector/models/` decode green/red at the same indices (0/1) if substituted — the extra magenta class is simply ignored by this map. **Lineage:** the 3-class detector line (with its leak-fixed per-class metrics, §09) and the venue-calibrated CIELab picker are retained in-repo as the measured evolution behind this stack.

Current validation status. The competition stack is committed and the components are individually exercised, but the final timed Round 2 track test has not yet been completed. This journal describes the implemented architecture and its component-level evidence without inventing a final race result.

09 Vision Iteration & Data-Driven Decisions

The final perception method was chosen after multiple approaches were measured and rejected



Live detection on the Raspberry Pi 5 — dual-pillar frame with class + confidence overlay (bench setup in docs/img/)

Detector evolution

Iteration	Evidence	Engineering response
HSV / connected components	Very fast (~0.43 ms/frame), but failed with both missed pillars and false positives.	Could not be rescued by a downstream verifier because missed proposals never reach the verifier.
tiny_pillar (111K params)	Lower wrong-side rate, but 9.7% no-call and lower overall decision performance.	Rejected in favour of stronger decision coverage.
nanodet_lite (1.17M params)	Clean-split macro F1 0.898 and pass-side accuracy 0.941; beat tiny model on six of seven metrics.	Selected as the neural winner on validation data.
Field transfer check	Neural winner failed to transfer reliably to real field footage; repository had already documented the one-session-data limitation.	Per-venue calibrated CIELab picker adopted as the interim competition stack; failed neural path preserved as evidence.
Magenta extension	Close-range reflection bleed joined marker colour into a large blob.	Dedicated magenta class, stronger chroma floor and morphological OPEN; parking-marker telemetry added.
3-class YOLO26n-320 v2	Retrained 19 Aug on the leakage-fixed, group-wise split, 80 epochs: mAP50-95 green 0.859 / red 0.792 / magenta 0.820.	Adopted for the 3-class detector line; weights (.onnx + .pt) and harness versioned in-repo.
Grid / pause-pass competition stack	2-class green/red ONNX with 5-of-7 vote + spatial lock; 7-column offset LUT with closeness bands; staged PAUSE→PASS→SIDE→YAW-BACK pass paired with the PID firmware. An earlier on-robot LUT peaked below full steering lock and let the body clip cubes — the offset floor is now the full 35°.	Committed as the flashed Nationals configuration (src/Round 2/competition/, §08); earlier stacks retained as evidence.

Validation integrity: the split was rebuilt after leakage was found

When validation numbers looked better than real behaviour, the team audited the dataset and found 25 duplicate image stems across the split boundary and 27 of 29 validation time buckets overlapping training. Instead of

adjusting the results, all previous numbers were withdrawn. A script then rebuilt a group-wise 473/124 split by capture bucket, and the models were retrained and re-measured. This was one of the most important systems-thinking decisions because it changed which evidence could be trusted — and it is why the v2 per-class numbers above are reported against the fixed split only.

Why venue calibration was adopted (and kept for the fallback)

Measured colour centres shifted substantially between capture sessions: the repository reports a 24.0-unit movement in the fitted red Lab chroma centre between two sessions, larger than the normal tolerance budget. Condition-matched calibration therefore became a procedure: calibrate in the actual venue light during check time, save the calibration and reuse it headlessly for the run. This converts lighting from an uncontrolled assumption into a managed setup step — and the same discipline applies to the competition detector: verify it on the actual field in venue light before trusting bench numbers.

10 Criterion 4 — Systems Thinking & Engineering Decisions

The robot was treated as one interacting system under constraints, not as separate parts

Primary constraints

Constraint	Design response
300 × 200 × 300 mm envelope	Compact stacked electronics and narrow chassis; camera and sensor mounts integrated into the frame.
≤ 1.5 kg total mass	Final vehicle is 785 g, leaving 715 g margin for compliance and limiting drivetrain/steering load.
Ackermann / non-skid steering	One steered front axle with MG90 linkage; one mechanically connected rear drive axle through a differential (a differential gear on the driven axle is rule-legal; differential-drive steering is not used).
Full autonomy	Two-button procedure: main power switch followed by the dedicated start button; no manual control during a run.
No wireless during rounds (rule 11.10)	Pi radios disabled for every round; the competition stack opens no network paths — preview is a local window only and headless by default.
Vision compute budget	Raspberry Pi 5 CPU-only; motion remains on ESP32 so perception load cannot block control.
Asymmetric scoring risk	The pass-side system is tuned to avoid wrong-side decisions; an uncertain/no-call is treated differently from a confidently wrong call.
Time before competition	Engineering evidence was versioned continuously in GitHub so design changes and failures were not lost.

Representative trade-offs

Choice	Alternative	Why the final choice won
Rigid LEGO redesign	Full printed chassis rebuild	Testing showed stiffness was the root issue; the lighter/cheaper LEGO redesign fixed wheel retention without a complete material change.
BNO055 absolute heading	MPU6050 gyro integration	Absolute/fused yaw avoids accumulating turn-rate drift over three laps.
TF-Luna LiDAR	HC-SR04 ultrasonic	Narrow beam produces a sharp opening edge, improving corner event timing.
Two processors	Single-board all-in-one compute	Vision delays cannot pause heading hold and steering.
P-only steering (R1) / full PID (R2)	One controller for both rounds	Round 1's stepped target rejects I and D (§04); Round 2's sustained pass offsets and re-join arcs justify clamped I and D. Each round runs the law its disturbance profile earns.
Staged grid-LUT pass strategy	Single-threshold swerve / earlier stacks	The pass is decomposed into measurable stages (400 ms pause at 45 px, edge-locked pass at full 35°, LiDAR-gated SIDE, fixed 28° yaw-back), each with its own tunable — failures localise to a stage instead of one opaque gain. Earlier stacks kept in-repo as the measured evolution.
Keep 20T:28T external stage	Regear for more speed	Lap arithmetic (§03) shows the ratio is not the binding constraint on round time; control and cornering matter more than an unloaded maximum.

Engineering principle used throughout. We chose X instead of Y when tests or constraints gave a reason. When an earlier decision was wrong — the steering sign, the leaky split, the first neural detector — the repository keeps the original reasoning and the reversal so that the engineering process remains auditable.

11 Risk Register & Iteration Cycles

Known failure modes were documented with mitigations rather than hidden

Risk	Impact	Mitigation / present status
Lighting differs from calibration session	Wrong colour / pass-side decision	Verify the detector on the actual field in venue light before the run; the CIELab picker line retains a saved-calibration procedure as documented evidence of the same discipline.
Pillar briefly blocks side LiDAR	False corner trigger	Round 2 corners trigger on the front LiDAR (20 cm) with a 1 s cooldown, so a pillar occluding a side sensor cannot fire a turn; Round 1 uses a rising edge + 1 s lidar mute.
Vision process stalls	Obstacle logic unavailable	ESP32 motion loop remains independent; robot continues on the last valid heading target.
Frame flex loosens wheels	Mechanical failure / no run	Resolved by rigid LEGO space-frame redesign; later sustained runs had no wheel loss.
Servo powered through logic rail	Voltage disturbance / reset risk	Servo now uses dedicated 5 V rail with common ground; ESP32 provides signal only.
I ² C address collision among TF-Lunas	Sensors cannot be distinguished	PCA9548A isolates each 0x10 TF-Luna on a separate channel.
Wireless active during a round (rule 11.10)	Rule violation	Procedure: disable Pi Wi-Fi/Bluetooth before every round; the competition stack opens no network paths — preview is a local window, headless by default.
Round 2 final integration incomplete	Unknown competition reliability	Final timed Round 2 test scheduled as the critical pre-competition validation task.
End-of-run parking not validated	Lost parking points / possible limiter contact	Parking-lot exit S-curve is in the competition firmware; the park-in shimmy controller (branch feat/round2-park) is written but not yet run — validate conservatively or scope out.

Iteration pattern used by the team

- Identify the observed failure or constraint.
- Record the failure before changing the design.
- Form a specific hypothesis about the cause.
- Change one subsystem or procedure that addresses the cause.
- Repeat the relevant test and compare with the earlier evidence.
- Keep rejected approaches in the repository when they explain a trade-off or prevent the team from repeating the same mistake.

12 Criterion 5 — Reproducibility & GitHub Quality

Another team should be able to understand, rebuild and run the system from the repository

Repository structure

Path	Purpose
docs/	Five WRO criterion documents, predicted power budget, testing workflow, evaluation raw data, figures and photos.
src/Round 1/	ESP32 Open Challenge firmware: heading hold, rising-edge corner detection, steering and drive logic (+ ultrasonic prototype, legacy IMU test).
src/Round 2/	competition/ — the flashed Nationals stack (test.cpp + detector.py + detect.py, with its own README); plus the 3-class ONNX detector line (weights and .pt in detector/models/), Pi sender, earlier ESP32 obstacle controllers, evaluation harness, vision A/B tools, and preserved lineages under history/.
models/	Team CAD: printed-part STLs (motor clamp, servo horn beam, TF-Luna enclosure), LEGO chassis capture (.lxfml) and build instructions.
schemes/	Signal and power wiring diagrams (v0.2, PNG + PDF, generated by a committed script) and the as-built circuit photograph.
t-photos/ · v-photos/	Team photographs (official + informal); vehicle from all six views plus detail shots.
video/	Required autonomous performance video links (one per challenge, ≥30 s each).
other/	Datasheets, setup notes and engineering test footage (test runs of 8–10 Aug).
tools/	Host-side utilities: Pi dataset capture rig (wro_capture.py) and serial logging.

Software environment needed to reproduce the robot

Part	Environment
ESP32 firmware	Arduino IDE or PlatformIO; Adafruit BNO055, Adafruit Unified Sensor, Adafruit BusIO and ESP32Servo.
Raspberry Pi vision	Python 3.10–3.12; NumPy, OpenCV, onnxruntime , PyAV, pyserial, Pillow, SciPy and v4l2 utilities. Detector weights (.onnx + .pt) are versioned in-repo.
Sensor setup	Three TF-Lunas must be configured to I ² C mode before use; all remain at address 0x10 and are separated by the PCA9548A.
Per-venue calibration	Verify detector per-class recall on the actual field; for the fallback picker, run the colour calibration routine in venue lighting, save, and reuse the saved values headlessly.

Competition artifacts

Artifact	Required	Status
README > 5000 characters	Yes	Present (10.3k), with team roles, documentation index and version history
Source code for programmed components	Yes	Present, src/
Vehicle photos from multiple views	Yes	Present — six views + detail shots
Team photo	Yes	Present — formal and informal
Wiring information	Yes	Present in schemes/; power overview in §05

CAD / model / chassis rebuild information	Yes	Present — STLs, .lxfml, build instructions; renders in §03
Open Challenge video	Yes	https://youtu.be/PbOQJ54wr-o (33 s, uploaded 12 Aug)
Obstacle Challenge video	Yes	https://youtu.be/dBjqMQ-jRBA (100 s, uploaded 12 Aug)
Commit history	—	100+ commits with conventional messages, PR-based merges, release tags (vo.2.0, vo.3.0) and release notes

GitHub repository: <https://github.com/teddriveomo/wro-2026-future-engineers> (mirror kept byte-identical at github.com/comede547/wro-2026-future-engineers).

13 Testing & Final Pre-Competition Status

What has been demonstrated, what remains pending, and what should be frozen before 26 August

Validation item	Status on 24 Aug 2026	Evidence / next step
Final robot mass	DONE — 785 g	Below 1.5 kg limit by 715 g.
Open Challenge three laps	DONE — 24 s	Latest team-reported full three-lap result; cross-checked against the drivetrain model in §04.
Wheel retention / chassis rigidity	DONE	Post-redesign sustained runs recorded without wheel loss; final frame retained.
Main power switch	DONE — installed	Use together with one separate start button for the WRO start sequence.
Round 2 controller / perception	COMMITTED — competition stack	src/Round 2/competition/ (test.cpp + detector.py); final timed end-to-end track test still pending. Pre-flight: the model file next to detector.py must match ONNX_MODEL_PATH ("best_ncnn.onnx").
Final Round 2 timed test	PENDING	Run on competition field / equivalent mat and record success, time, obstacle-side errors and stop behaviour.
Parking	EXIT: IN FIRMWARE · PARK-IN: WRITTEN, NOT RUN	Boot S-curve exit is part of the competition firmware; park-in shimmy controller on feat/round2-park — validate conservatively before enabling.
Loaded straight-line speed	RE-MEASURE	Tape + timer over a marked distance before rounds; until then the journal carries only the derived 1.15 m/s unloaded ceiling (§03). The same run measures the driven lap line (§04).
Axle-load split	MEASURE	Front/rear weight split settles the traction-vs-stall regime (§03).
Radios off (rule 11.10)	PROCEDURE	Disable Pi Wi-Fi/Bluetooth before every round; verify before staging.

Recommended final test record for Round 2

- Run at least 5–10 full attempts with mixed red/green pillar arrangements.
- For every run record: completion yes/no, total time, wrong-side pillar events, moved pillars, wall contacts, number of recoveries/reverses and whether the robot stopped correctly after three laps.
- For parking record: bay detected yes/no, entry attempt yes/no, limiter touch yes/no, fully/partly parked and final orientation.
- Freeze the exact source revision used for the final successful run and commit the result table/video/log with a clear message.

Competition dates. 26–28 August 2026. The final Round 2 and parking tests are therefore the highest-priority remaining engineering tasks. Results should be added to the repository and this journal only after they are actually observed.

14 Evidence Map to the WRO 2026 Rubric

Where each judging criterion is supported in this journal and repository

WRO criterion	Evidence provided
Mobility & Mechanical Design	Final chassis and photos; Ackermann steering; MG90 range/centre; N2O + TB6612 drivetrain; 20T:28T ratio; wheel dimensions; final mass; torque/traction/acceleration derivation; CAD of printed parts ; failure-driven frame redesign; Open three-lap test.
Power & Sensor Architecture	As-built power tree + circuit photo; installed main switch; separated motor/servo/electronics paths; predicted power budget; 3× TF-Luna + PCA9548A; BNO055; camera; placement/beam-width reasoning with POV imagery; calibration and noise controls.
Software Architecture & Obstacle Strategy	Open flowchart; verbatim firmware logic excerpt; P-vs-PID control design rationale with the documented sign-fix ; 12-turn stop; Round 2 state machine; perception pipeline flowchart with verbatim Python decision code ; two-stack strategy with on-site selection criterion; explicit pending validation.
Systems Thinking & Engineering Decisions	Constraints incl. rule 11.10; alternatives table; dataset leakage audit and withdrawal of numbers; neural→classical→retrained-neural evolution; chassis stiffness diagnosis; risk register; measured reasons for decisions.
Reproducibility & GitHub Quality	Public repository + byte-identical mirror; role-attributed README; code incl. versioned weights; CAD/STLs; wiring; photos; required videos; environment; calibration setup; test workflow; 100+ commit history with PR merges and release tags.

Final team checklist

- ☒ Engineering story explains why decisions were made, not only what was assembled.
- ☒ Mechanical design, steering, drivetrain and torque budget are illustrated and reproducible.
- ☒ Power distribution and sensor placement are documented with imagery.
- ☒ Software control flow and obstacle strategy are understandable without reading raw source first — and the key source is excerpted verbatim.
- ☒ Multiple iteration cycles (chassis, steering sign, dataset split, detector lineage) are supported by test evidence.
- ☒ GitHub includes code, wiring, models/build information, photos, videos and meaningful history.
- ☐ Final Round 2 timed validation must still be added after it is performed.
- ☐ Parking validation must still be added after it is performed.
- ☐ Loaded speed + driven-line tape measurements before the competition copy is frozen.

Conclusion

The final TED Drive vehicle is the result of repeated engineering iteration rather than a single build. Mechanical flex changed the chassis, an inverted sign changed the steering law, data leakage changed the validation method, poor field transfer changed the vision architecture twice, and power/sensor integration changed the wiring. The final Open Challenge system has completed three laps in 24 seconds on the latest reported test — a figure this journal cross-checks against its own drivetrain model rather than asserting — and the complete vehicle weighs 785 g. The remaining work is clearly bounded: finish timed Round 2 validation, complete parking, and take the two single-session measurements (loaded speed, axle-load split) that close the mechanical chapter. Keeping these items explicit is part of the engineering process and makes the documentation reproducible and credible.

References

- World Robot Olympiad Association, *WRO Future Engineers 2026 — General Rules*, version 15 January 2026, especially Appendix C: Engineering Journal evaluation and the Documentation Rubric.
- Team TED Drive public GitHub repository: <https://github.com/teddriveomo/wro-2026-future-engineers>
- Open Challenge video: <https://youtu.be/PbOQJ54wr-o> · Obstacle Challenge video: <https://youtu.be/dBjqMQ-jRBA>

Journal status. Final pre-competition edition rev 3, prepared 24 August 2026 from the public repository at commit b715a4b, the WRO 2026 Appendix C rubric and the team's latest supplied measurements/photos. Pending test items are intentionally labelled as pending rather than presented as completed results.